Upper Confidence Bound (UCB) for Online Advertisement Selection

Executive Summary

Online advertising systems continuously face the challenge of deciding which advertisement to display to users. The goal is to maximize clicks, conversions, or revenue while simultaneously learning which advertisements perform best.

The **Upper Confidence Bound (UCB)** algorithm is a popular Reinforcement Learning and Multi-Armed Bandit approach that balances:

- **Exploitation**: Showing advertisements known to perform well.
- **Exploration**: Testing advertisements with limited historical data.

This report explains how UCB works, why initial exploration is important, and how it can be applied to a system that selects one advertisement from a set of ten advertisements.

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1. Problem Statement

Assume an advertising platform contains the following 10 advertisements:

```
Ad A
Ad B
Ad C
Ad D
Ad E
Ad F
Ad G
Ad H
Ad I
Ad J
```

For every incoming user:

1. Select exactly one ad.
2. Display the ad.
3. Observe the outcome:
   - Click = 1
   - No Click = 0
4. Update learning statistics.

The objective is to maximize the total number of clicks over time.

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2. The Exploration vs Exploitation Challenge

Suppose the true Click Through Rates (CTR) are:

Ad	True CTR
A	2%
B	3%
C	5%
D	1%
E	8%
F	4%
G	2%
H	6%
I	3%
J	1%

These values are unknown to the system.

Problem Scenario

Imagine the first user sees Ad A and clicks it.

The system now observes:

```
Ad A CTR = 100%
```

If the system immediately assumes Ad A is the best ad, it may continue showing Ad A repeatedly and never discover that Ad E actually has the highest CTR.

This creates a learning problem.

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3. UCB Solution

UCB assigns a score to every advertisement.

The score consists of two components:

Exploitation Component

Average observed reward:

```
Q(i)
```

This answers:

"How good has this ad been so far?"

Exploration Component

Confidence bonus:

```
c × √(ln(t)/N(i))
```

This answers:

"How uncertain am I about this ad?"

UCB Formula

$$UCB_i = Q_i + c\sqrt{\frac{\ln{(t)}}{N_i}}$$

Where:

Symbol	Meaning
Qi	Average reward of ad i
Ni	Number of times ad i has been shown
t	Total number of rounds
c	Exploration parameter

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4. Initial Exploration Strategy

A common production strategy is:

Show Every Ad Once

First 10 Rounds

```
Round 1  -> Ad A
Round 2  -> Ad B
Round 3  -> Ad C
Round 4  -> Ad D
Round 5  -> Ad E
Round 6  -> Ad F
Round 7  -> Ad G
Round 8  -> Ad H
Round 9  -> Ad I
Round 10 -> Ad J
```

Why?

Because UCB requires observations for every advertisement.

Without observations:

```
Ni = 0
```

and the formula becomes undefined.

This guarantees:

- Every ad receives initial traffic.
- Every ad obtains an initial CTR estimate.
- UCB starts with real data.

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5. Worked Example

Assume after the first ten impressions:

Ad	Clicks	Impressions	CTR (Q)
A	0	1	0.0
B	1	1	1.0
C	0	1	0.0
D	0	1	0.0
E	1	1	1.0
F	0	1	0.0
G	0	1	0.0
H	1	1	1.0
I	0	1	0.0
J	0	1	0.0

Total rounds: `t = 10` Assume: `c = 1`

For Ad B:

$$Q_B = 1$$
$$N_B = 1$$
$$\ln(10) = 2.3026$$

UCB score:

$$1 + \sqrt{2.3026}$$
$$1 + 1.517$$
$$2.517$$

For Ad A:

$$0 + 1.517$$
$$1.517$$

Since Ad B has a larger UCB score, it is more likely to be selected.

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6. How Learning Evolves

Over time:

High Performing Ads

```
Higher CTR
More clicks
More impressions
```

Examples:

```
Ad E
Ad H
```

These ads receive more traffic.

Low Performing Ads

```
Lower CTR
Fewer clicks
```

Examples:

```
Ad D
Ad J
```

These gradually receive less traffic.

New or Uncertain Ads

If an ad has been shown only a few times:

```
Ni is small
```

The exploration bonus becomes larger.

UCB may decide:

"I need more information about this ad before ruling it out."

This prevents missing potentially valuable advertisements.

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7. Traffic Allocation Diagram

```
              START
                |
                v
       +-----------------+
       | User Arrives    |
       +-----------------+
                |
                v
       +-----------------+
       | Compute UCB     |
       | For All Ads     |
       +-----------------+
                |
                v
       +-----------------+
       | Select Highest  |
       | UCB Score       |
       +-----------------+
                |
                v
       +-----------------+
       | Show Ad         |
       +-----------------+
                |
                v
       +-----------------+
       | Observe Click   |
       | or No Click     |
       +-----------------+
                |
                v
       +-----------------+
       | Update CTR and  |
       | Impression Cnt  |
       +-----------------+
                |
                |
           Next User
```

8. Optimistic Initialization Alternative

Instead of showing every ad once, some systems initialize:

```
Qi = 1.0
Ni = 1
```

for every advertisement.

This assumes:

"Every ad is excellent until proven otherwise."

Initial state:

Ad	Initial Q
A	1.0
B	1.0
C	1.0
D	1.0
E	1.0
F	1.0
G	1.0
H	1.0
I	1.0
J	1.0

As observations arrive, poor ads quickly lose their optimistic estimates.

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9. Recommended Production Approach

For advertisement serving systems:

Step 1

Show each ad at least once.

Step 2

Maintain:

```
Clicks
Impressions
CTR
```

Step 3

Compute UCB scores.

Step 4

Display the ad with the highest UCB score.

Step 5

Update statistics after every impression.

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10. Key Takeaways

✓ UCB balances exploration and exploitation automatically.

✓ Every advertisement should receive initial exposure.

✓ Showing each ad once is the simplest and most common initialization strategy.

✓ Optimistic initialization is valid but less commonly used with UCB.

✓ Over time, UCB naturally shifts traffic toward better-performing advertisements while continuing to gather information about uncertain ads.

✓ This makes UCB an effective algorithm for online ad optimization, recommendation systems, and A/B testing environments.

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